Cayleb Alvarez-James

AI Agent Systems Engineer - Autonomous Agents, Agent Harnesses & Self-Improving Infrastructure

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Summary

AI agent systems engineer and technical founder with a year in agentic systems and harness engineering, beginning while finishing high school. I work across several harnesses instead of just one: OpenCode/DeepSeek Harness, Claude Desktop and Duet, Gemini Antigravity, Freebuff, ChatGPT Codex and Claude Code - routing work by available usage and fit; the current centre is Muse, Claude Duet and the DSH Fleet I am finishing. Three systems define the work: Ergon (self-configuring employee agents), the DSH Fleet (distributed, self-improving dynamic-swarm orchestration) and OpenBell (tiny models built and evaluated in weight space). 1,200+ GitHub contributions in the last year; 4,000+ automated tests across five shipped systems.

Experience

Founder - Independent AI Agent Systems | Oct 2025-present

Architected Ergon, the DSH Fleet and OpenBell: self-configuring agents, dynamic-swarm infrastructure, weight-space model evaluation.

Selected Projects

Ergon - Self-Configuring Employee Agents & Autonomous Role Learning

Node.js, C#, Python, SQLite

Ergon case study

- Core loop: install, observe recurring work and tools, ask targeted questions, build a persistent role/task model, propose and build a capability, evaluate and review, earn the grant. Personalisation is a harness primitive: per-employee memory, persona, skills and tools evolve with the job.
- Proven skills enter a shared library others earn through the same gate; permissions stay reversible.
- Standalone product in development: signed Windows installer and taskbar app, demo connectors and ticket-scoped remote repair. RHL: a proposed first deployment - planned, not implemented. 622 tests pass.

DSH Fleet - Distributed, Self-Improving Agent Infrastructure

Python, TypeScript

DSH Fleet case study

- Agent control plane spanning desktop, Windows VM, phone VM and browser surfaces. Slack is the open coordination layer: my agents and bring-your-own external agents share one ticket/context/outcome protocol. A lead forms a task-shaped swarm per objective - no standing bench - dissolving after one auditable outcome.
- The Omni Loop - my recursive self-improvement architecture - is every swarm's iteration engine (attempt, measure, rewrite; 3 iterations max per approach, then wiped and replaced) and is applied to itself after every ticket, gated: a candidate ships only on a fresh checker's measured win on a fixed scorecard (checker-pass rate, cost per checker-passed task, frozen regression set); any regression rejects it; improvements are versioned, reversible; it never edits its own gates.
- Ladder inside every swarm: harness preferences first; where none covers an area, a skill created or updated within the last 30 days is sourced before acting; raw model knowledge last; reusable work packaged back into skills. Design is guided by multi-agent research (Anthropic/OpenAI systems work, MAST failure taxonomy, Elephant-Goldfish spec review, good-views verification, cost-per-passed-task) - guidance, not a cage. The fleet has caught its own swarms filing false 'done' reports, each found by re-checking the primary source.

OpenBell - Weight-Space Training & Evaluation for Trading Models

Python, Rust

OpenBell case study

- Direct weight-space construction: low-dimensional weights for tiny specialist models, not prompted strategies. Weights and predictions are frozen before scoring: inspectable, falsifiable.
- Evaluation is deliberately adversarial: chronological splits, transaction costs, negative results retained, pre-registered kill rules, no money test until training clears the exit-geometry breakeven. Conventional distillation failed on both benchmark instruments; directly authored weights produced the strongest results, including one Nasdaq configuration net-positive on historical data.
- Kalshi, futures and meme desks provide live, capital-isolated proving grounds: parallel slots and 2x ladder runs under reconciled real-market accounting, one experiment never contaminating another.

Sophos - Coding-Agent Desktop App

React, TypeScript, Tauri

github.com/cayleb-james2008/sophos

- Coding-agent desktop app from saved profiles; sessions and transcripts survive restarts.
- 1,224 interface tests, 82 Node tests and 56/56 Windows checks pass; merged after full hosted CI.

dotz - AI Coding Agent Dashboard

Rust, Tauri, JavaScript, SQLite

github.com/cayleb-james2008/dotz

- Desktop app running and supervising AI coding agents: model choice, safe process-tree cleanup.
- 828+ core tests pass on each of Windows, macOS and Linux; all 6 hosted check jobs pass; merged.

Asmodeus - Trading Research on Past Data

Rust

Asmodeus case study

- Tests trading ideas on historical data in strict time order, with fees and safeguards against peeking ahead; rejected ideas are logged. 1,111 Rust tests pass. Research only - no real money.

Industry AI Suite - Source-Checked Review Tools

Python

github.com/cayleb-james2008/industry-ai-suite | Review history implementation

- Ten review tools; every finding shows its source and a person decides. 382 Python tests pass.

Skills

AI & agents: Multi-harness agent operations (OpenCode/DeepSeek Harness, Claude Desktop and Duet, Gemini Antigravity, Freebuff, ChatGPT Codex, Claude Code), autonomous role learning, dynamic-swarm orchestration, recursive self-improvement systems (the Omni Loop), evaluation design, earned permissions and approval gates

Languages: Python, TypeScript, JavaScript, Rust, HTML/CSS

Models & data: Weight-space model construction, linear/LDA specialists, chronological backtesting, experiment design, React, Tauri, SQLite, web APIs, Git/GitHub, Linux

Additional Experience

Robert H. Lord Company (RHL) - AI Concept Discussion & Warehouse-Board Demo | Sep 2026-present

David Cooke Plaster Co. - Plaster Work | Summer 2026

Dunkin' | 2024-2026

Tommy's Pizza | Summer 2024

Education

Portland High School | Portland, Connecticut | Graduated 2026

Coursework: Cybersecurity; AP Computer Science